

Milestone 2

Section 1: Engineering Requirements

In preparing the crew for the mission, the years of preparation must include preparing for worst case scenarios. Training the crew to be able to handle dust storms, extreme temperature fluctuations, delayed communication to earth, and extreme difference in gravity to fully prepare crew members to successfully acclimate mars.

Next, we must time the dates correctly with the perfect time it takes for mars and earth to align for the maximum optimization of efficiency and success of said mission. It must correlate within the 26-month time frame that Earth and Mars align. This is a must, if missed calculations for a trip we won't have another opportunity for two years. Along with timing the blast off correctly, the crew must be prepared to be en route from a range of 4 to 11 months depending on the path of transportation chosen.

We must have satellites in orbit in earth's orbit as well as mars orbit to be able to establish a form of communication. We must also prepare cargo ship to launch alongside but keep it in mars orbit to be able to have our supplies at the ready for expansion on mars. Said supplies must have a protective coating, especially the supplies that will be used for crew quarters, or any framework that will be housing supplies that crew needs, and coating must be able to withstand dust storms as well as extreme temperature fluctuations.

Speaking of extreme weather fluctuations, crew members clothes as well as materials that will be exposed to mars air does need proper temperature control functions, Mars temperatures fluctuate between 20 degrees Celsius to as low -195 degrees Celsius during the polar winters. Not only are the temperatures so extreme, but Mars' duration of seasons lasts twice as long as they do on earth.

When it comes to landing successfully on Mars, a protective heat shield must be used as entering the mars atmosphere can get extremely hot. The atmosphere on mars also doesn't lend itself to a seamless landing, so parachutes and rockets must be equipped to ensure a successful landing onto mars surface.

Once on Mars, rovers that will be on mars to help with human expansion should be equipped with drills to be able to harvest the ice on mars as well as the regolith to be able to use those resources in the human exploration of mars.

Section 2: Design Constraints and Assumptions

Creating an engineering design may be very interesting, but it has its own limitations and constraints. These consist of things like budget limitations, materials we will be using and whether they are available or not, moreover if we will be capable of using these to manufacture things according to our needs, and if we do prove to be capable would the environment at Mars let us do what we need to? It is better to consider these constraints early during the design process rather than when it is too late.

One of the most important constraints, something that we cannot look over, is the budget limitation. We have a budget limit of fifty billion Dollars – this budget needs to be utilized in a way that fits all design requirements excluding the journey to and from Earth. It must cover the manufacturing and transportation using cargo deliveries, surface vehicles, and construction. Additionally, the development of the outpost and the on-board computing systems along with using surface/orbital antennas for a stable communication system with the Earth; and most importantly the establishment of a stable and reliable power system with multiple storage sources in case of emergencies. Furthermore, the cost to build living quarters, radiation protection, and regolith shields must also be covered along with basic human necessities such as food production, waste management, water processing, and recycling.

Another constraint would be the establishment of a power system, the power systems engineer would have to figure out where the energy will be coming from whether it will be solar energy or another form, as of right now it is assumed that the energy production system would be a nuclear fission reactor but it is to be decided how they are going to execute their ideas. Along with that, they would also have to figure out a way to distribute power into different subsystems of the design outpost while remaining within the budget limit; the use of energy grids is the main idea for energy distribution. An efficient source of power storage must be found for times where power generation is reduced so that none of the research equipment turns off. Other than the main energy generation method, a secondary or backup method must be used so that the critical systems do not shut down, and the astronauts do not lose any important data.

The loss of data brings us to our third constraint which is communications and computing systems which will be controlled by the communications engineer who has the responsibility to maintain a stable connection with earth using antennas, communication relays, and data transmission. The delay between Earth and Mars cannot be dismissed but in case of emergencies they would need to establish on board computing requirements for emergencies where waiting on an answer from Earth would not make sense. They will also need to create computing requirements so that all the data that is being collected is first saved in their own devices before it is sent out to the Earth.

Further constraints include the schedule limitation; the project is intended to be of five years and the astronauts need to fulfill their mission within that time limit hence, technologies that use extensive development and testing may not be used and rather use the technology that is more

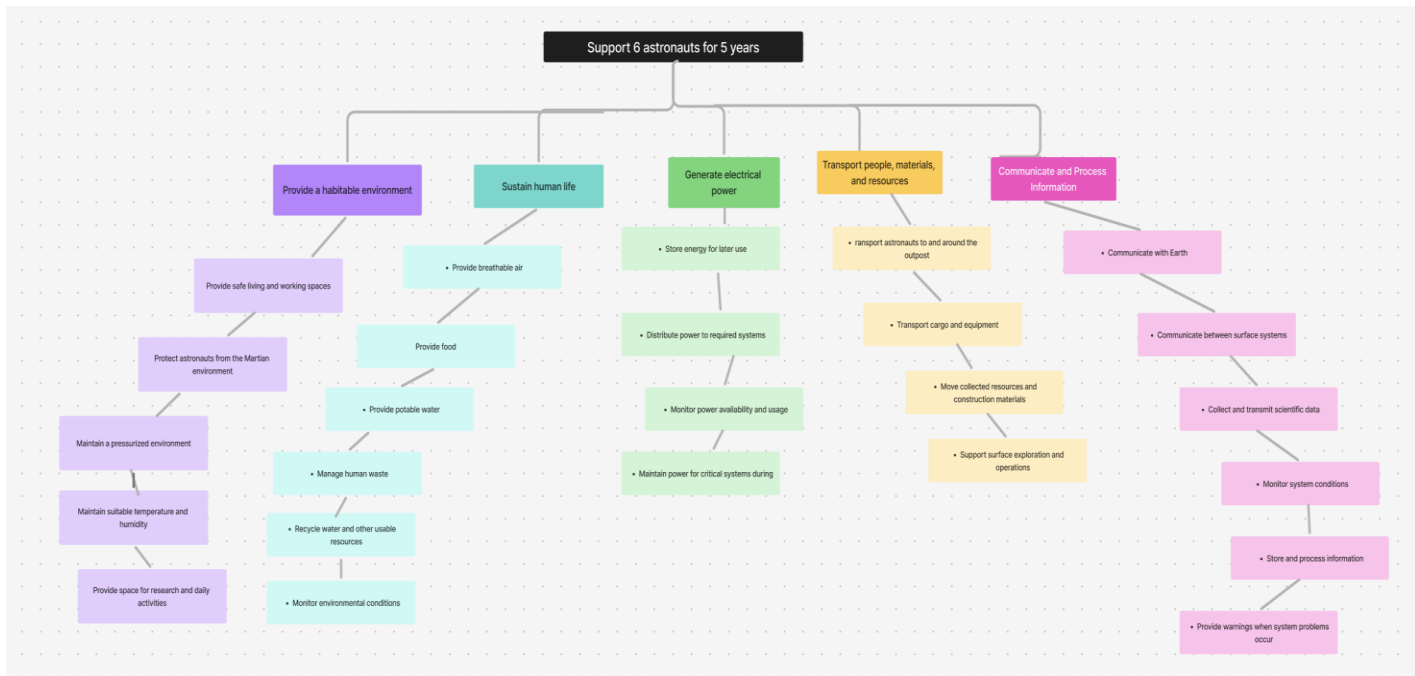
reliable and stable to save more time. The outpost must be designed so that the atmosphere on Mars does not affect it as Mars has extreme temperatures, dust storms, low atmospheric pressure, and is exposed to radiation; the astronauts must decide on their manufactural materials in a way that they are not affected by these extreme conditions. Another big limitation would be to provide all six astronauts with constant life support. This will consist of the life support engineer to establish a way of oxygen production, water recycling, etc.

Finally, the mission duration is expected to be of five years and we are assuming that all our system requirements including, power systems, communication, transportation, safety, life support and manufacturing will all be covered under the budget of fifty billion dollars and will be designed in a way that it supports long periods of time on Mars. It is also assumed that there will be constant and stable communication between the Earth and Mars despite the communication delay that is expected between the two planets. Along with that, all human necessities will be provided for, and the astronauts will not run out of anything for the 5 years they are spending on Mars.

The overall function of the Mars Research Outpost is to support six astronauts for a five-year mission while enabling scientific research and technology development on the Martian surface. To accomplish this overall mission, the system is divided into several primary functions: sustaining human life, providing and managing power, protecting the crew and equipment, enabling communication and information processing, supporting scientific research, supporting Mars surface operations, and maintaining and expanding the outpost. Each primary function is further divided into smaller functions that describe what the system must accomplish without specifying the specific technologies or components that will be used.

Each function represents an activity the outpost must perform rather than a specific technology. For example, the system must provide electrical power, but the functional decomposition does not require selecting solar panels, nuclear reactors, or another power source as we are still testing different ideas to decide which is most efficient and inside our budget constraints.

Section 3: Functional Decomposition Diagram:



Section 4: Preliminary Subsystem Identification

Subsystem 1: Habitat and Structure

Primary function: The astronaut's habitat will remain in the atmosphere pressure and temperature control of safe human limits for all 6 members.

Key Inputs: Structural Materials, Mass constraints, Martian Regolith.

Key Outputs: The Martian regolith will help defend against radiation. Structural Material will help build a dust tolerant exterior and thermal control.

Relationship to Other Subsystems:

Requires electricity from the power systems to control the pressure and

maintain a safe temperature, and to recharge all research equipment. Works with the communications and computing system to store and transmit all the data gathered.

Contribution to Overall System: The design concept is a lightweight inflatable module and will be deployed on the surface. There will be multiple meters placed around the Martian regolith by robotic rovers. Adding 3D printed elements and laser vitreous multi-material transformation will melt and solidify the regolith into a ceramic. These can be used for green houses, connecting tunnels, launch pads, and future expansion.

Subsystem 2: Life Support

Primary Function: Maintain the environmental conditions necessary for human survival.

Key Inputs: Water, energy, food supplies, atmospheric resources, waste

Key Outputs: Breathable air, usable water, food, processed waste, environmental control

Relationship to Other Subsystems:

Depends on Power and Habitat and interacts with Construction & Manufacturing and Safety.

Contribution to Overall System: Keeps six astronauts alive and supports the five-year mission.

Subsystem 3: Power Systems

Primary Function: To generate as much power as possible while also storing it for backup power. To distribute power efficiently so that the research equipment does not get affected by reduced power.

Key Inputs: Sunlight, Nuclear fuel, Building material for nuclear reactors and solar panels, batteries.

Key Outputs: Stable electrical power to all subsystems, stored energy, data and information about power generation and consumption during the five years.

Relationship to Other Subsystems:
Provides energy to life support for oxygen production, recycling water.
Provides energy to the on-board computing and communication systems.
Distributes energy to every subsystem for research equipment all over the outpost.
Provides back up power to critical systems.

Contribution to Overall System:
Provides electrical power to the entire outpost so that every subsystem can utilize it for its uses.

Subsystem 4: Communication and Computing

Primary Function: Provide stable communication between Earth and Mars.
Send and receive data from Earth.
Store all the data before it is sent to Earth.

Communicate with all the rovers and other crewmates.

Key Inputs: Data from all the research equipment used including rovers, power supply to keep communication and computers running, Communication signals from Earth.

Key Outputs: Data is sent out to earth and received by the astronauts on Mars. All the data is processed and saved. Notifies any problems with the system or the equipment. Monitors all the subsystems like life support and power systems so that any problems are detected.

Relationship to Other Subsystems:

Relies on Power Systems for all its electrical requirements. Collect all the information from all subsystems like life support, safety and risk, habitat and store it. Fulfills all the computing requirements for any robotic equipment such as rovers.

Contribution to Overall System: Acts as the outpost control system, detect any faults in all the subsystems, keep all the data stored and safe while also providing a stable and reliable connection with Earth.

Subsystem 5: Transportation

Primary function: Provide a reliable form of transportation for crew and cargo units.

Key Inputs: Electrical power supplied by power systems, route and navigating

systems. Communication connection between Earth and main spacecraft.

Key Outputs: Transported crew members, samples, tools and equipment necessary to establish outpost. Safe return of crew once mission is complete.

Relationship to Other Subsystems:

Relies on power systems to recharge batteries and equipment needed from transportation through arrival. Uses communications and computing for navigation in real-time, tracking, and safety of routes chosen. Relies on life support systems for oxygen distribution and temperature control.

Contribution to Overall System: Allows for exploration to outpost successfully to establish life on Mars. Crew will be able to explore and document, test samples, and bring necessary equipment.

Subsystem 6: Construction and Manufacturing

Primary function: Provide tools, equipment, and materials needed to assemble outpost infrastructure on Mars.

Key Inputs: Electrical power supplied by power systems for operation tools and machinery needed to construct outposts on Mars. Regolith and Martian materials as local materials for 3D printing or other necessities. Structural design from project management systems.

Key Outputs: Habitat living spaces, shielding and protection from radiation, external structures. Fabricated parts and

structural components. The ability to provide maintenance once established outpost on mars.

Relationship to Other Subsystems:

Relies on power systems for electricity needed for equipment, 3D printing systems, and machinery. Using Transportation through rovers to move materials gather regolith, and fabricated parts to building sites. Communication and Computing systems needed for design of structure and blueprints. Safety and Risk management needed to ensure safe and proper precautions taken for human survival in structures.

Contribution to Overall System: Allows for outpost growth overtime using recourses such as regolith and other Martian materials for sustainability of crew for duration of mission.

Subsystem 7: Safety And Risk

Primary function: All critical systems will have fault detection, isolation, and extreme recovery capability.

Key Inputs: Hazard identification; system reviews of every subsystem, operation phases, and environment factors.

Key Outputs: Safety of the astronauts from medical injury, illness, or even psychological stress, there will be medical kits, exercise equipment, and scheduled psychological appointments

Relationship to Other Subsystems:

Tracks risk from every subsystem and shows residual after every mitigation

Contribution to Overall System:

Mitigates every major threat, enables mission success and reliability, reducing the probability of fault detection, supporting long duration and expandable operations.

Subsystem 8: Systems Engineering/Project Manager

Primary function: Make sure the project is well coordinated, and all the subsystems are working together.

Track the project's progress and make sure the project remains within the set budget of fifty billion dollars.

Key Inputs: Information about the project requirements and data from all the subsystems such as designs from power, communication, transport, and habitat, etc.

The schedule timeline is also needed to make sure everything is done according to the timeline.

Key Outputs: The project is built within the time limit of five years. All subsystems work together efficiently. The budget is not exceeded, and the project remains within the budget.

Relationship to Other Subsystems: All subsystems report to the Project manager for their budget usage.

Contribution to Overall System: Keeps all subsystems coordinated so that all systems work efficiently.

Section 5: Preliminary System Architecture Block Diagram

